



OST
Ostschweizer
Fachhochschule

Preprocessing (2) & (3)

Data Analytics

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Studiengang Informatik

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Administratives / Course Overview

Course Syllabus «Data Science»

1	Introduction	Chapter 1
	Preprocessing & Visualization – Part 1 (data objects, basic statistics)	Chapter 2
2	Preprocessing & Visualization – Part 2 (data cleaning, correlation)	Chapter 3
	Preprocessing & Visualization – Part 3 (data transformation, discretization)	Chapter 3
3	Association Analysis	Chapter 6
	Classification – Part 1 (basic concept, naive bayes, evaluation)	Chapter 8
4	Classification – Part 2 (decision trees)	Chapter 8
	Classification – Part 3 (ensemble methods)	Chapter 8
5	Clustering – Part 1 (basic concept, partitioning methods, measuring similarity)	Chapter 10
	Clustering – Part 2 (hierarchical methods)	Chapter 10
6	Clustering – Part 3 (density-based methods)	Chapter 10
	Clustering – Part 4 (outlook on advanced methods, evaluation)	Chapter 10

Agenda

What we will cover today...

- **Dealing with Noisy and Missing Data**
- Redundancy and Correlation Analysis
- Normalization
- Discretization
- Data Reduction



Dealing with Noisy and Missing Data

Why does Missing Data Happen? Why do we have to care?

- Data is not always available
 - e.g., many tuples have no recorded value for several attributes, such as customer income in sales data
- Missing data may also be due to
 - Missing data may be due to equipment malfunction
 - Inconsistent with other recorded data and thus deleted
 - Data not entered due to misunderstanding
 - Certain data may not be considered important at the time of entry
 - No registered history or changes of the data
- How much of a problem are missing values for further analysis? There are two main issues...
 - **Loss of efficiency** → because information is missing
 - **Introduction of bias** → i.e. if the missing data does not occur randomly

Dealing with Noisy and Missing Data

Types of Missing Data

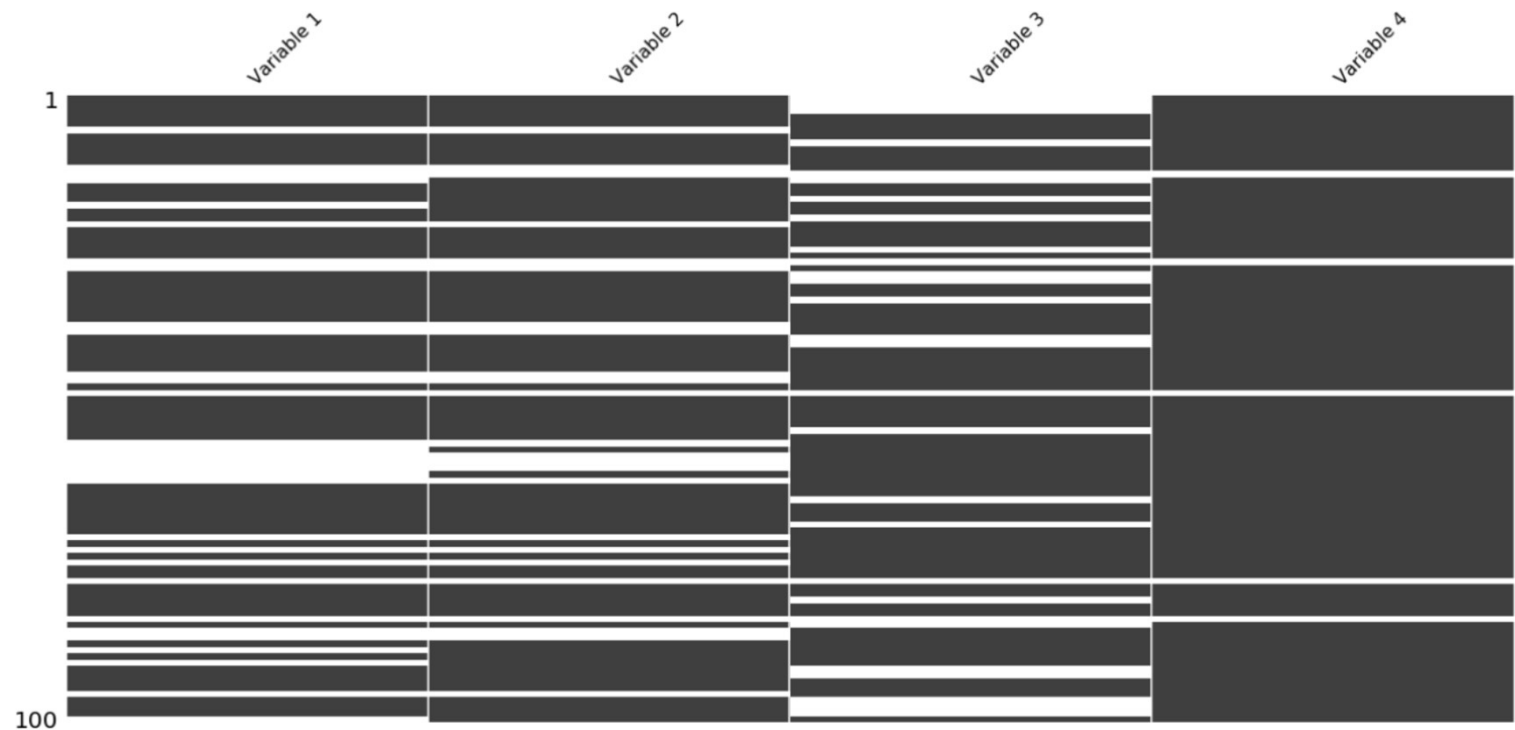
- Missing completely at Random (MCAR)
 - If the missing values do not depend on the values of variables in the dataset
 - The probability of a value being missed is only related to the underlying distribution parameters, but neither to the missing nor the observed values.
- Missing at Random (MAR)
 - If the probability that values are missing depends only on the set of observed values, but is completely unrelated to the missing values that should have been obtained.
 - In other words, the occurrence of missing values can be explained by the observed values.
- Missing not at Random (MNAR)
 - If the probability that responses are missing is related to the true value of the responses (in addition to the ones actually obtained)

Sources:
Little, R., A test of missing completely at random for multivariate data with missing values. *Journal of the American Statistical Association*, 1988

Dealing with Noisy and Missing Data

Visual representation of a Missing Value Pattern

- Black means observed, white means missing.
- Image created in Python with random data and the Missingno Package from Bilogur.



Sources:
Bilogur, A. (2018). Missingno: A missing data visualization suite. *The Journal of Open Source Software*, 3 (22), 547

Dealing with Noisy and Missing Data

Dealing with Missing Values

- Discard measurements with missing values in their attributes
- Fill missing values with estimated ones
 - Fill in the missing value manually which requires a lot of time and background knowledge
 - Use a global constant to fill the missing value e.g. “unknown”
 - Use a measure of Central Tendencies (e.g. mean, median, mode...)
 - Regression
- Maximum likelihood procedures
 - Maximum likelihood is a method, where based on a number of observations, the distribution parameters are estimated so that the observed data has the highest probability of being part of the evaluated distribution.
 - The concept of this estimation can also be used to create a statistical model that might have created missing values. Such a model would help to impute the missing values in a second step.
- Latest Approach: Machine Learning Based Methods

Sources:
Garca, S., Luengo, J., & Herrera, F. Data preprocessing in data mining. Intelligent Systems Reference Library. Springer, 2015

Dealing with Noisy and Missing Data

Noisy Data (Attribute Noise)

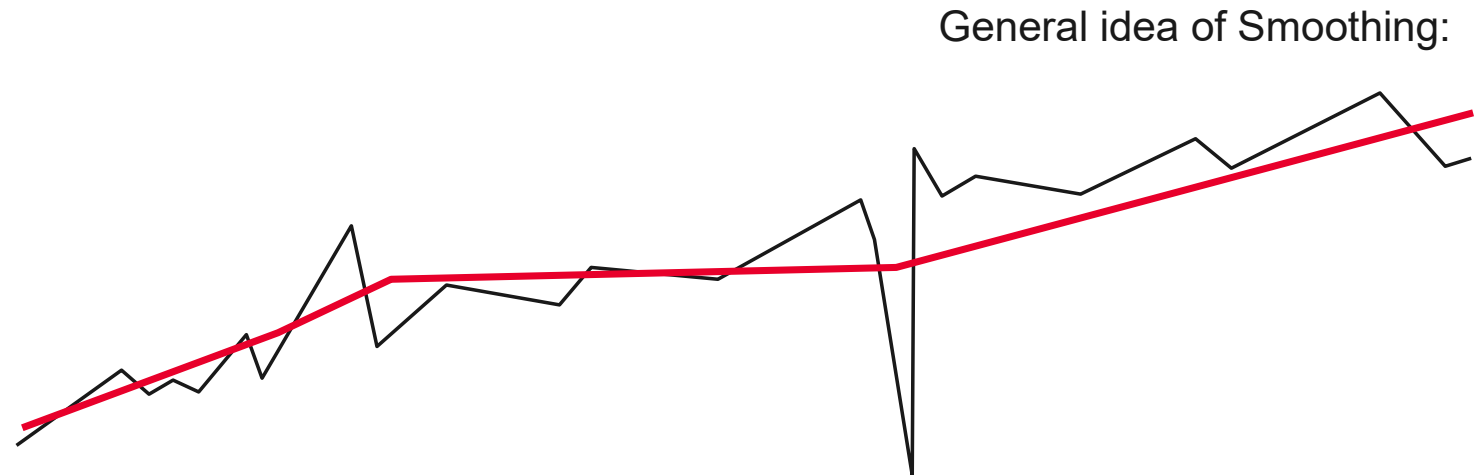
- When real world data is collected, there might be some disruption within this data, called noise
 - Real world data consists of a true signal and noise, which is a random error or variance
 - It's corrupted or distorted data that is mostly meaningless and thus not useful for analysis (only signal)
 - Noise might interfere with the interpretation of models or affect the performance of algorithms negatively
- Removing Noise is sometimes nearly impossible and mostly very time consuming
 - Optimal solution would be methods that are robust" against noise i.e. that a are capable to build models that are insensitive to data corruptions and suffer less from impact of noise
- Other Approaches
 - Filters detect and eliminate noisy instances
 - Regression – Smooth by fitting the data to (multiple) linear regression
 - Smoothing by bins (next slide)

Sources:
Garca, S., Luengo, J., & Herrera, F. Data preprocessing in data mining. Intelligent Systems Reference Library. Springer, 2015

Dealing with Noisy and Missing Data

Binning Methods for Data Smoothing

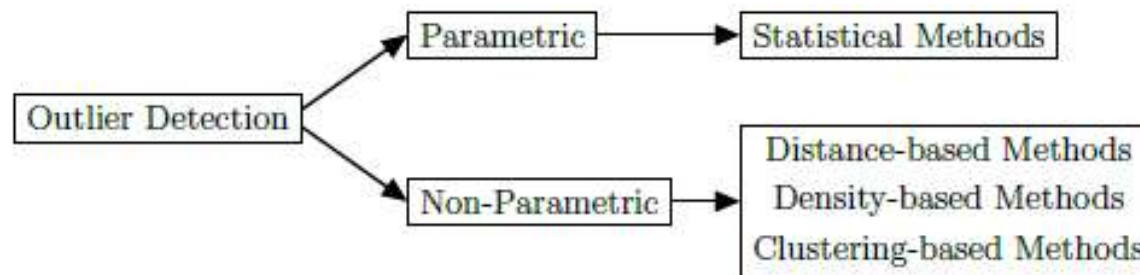
- Sorted data for price (in CHF): 4, 8, 15, 21, 21, 24, 25, 28, 34
- Partition into (equi-depth) bins:
 - Bin 1: 4, 8, 15
 - Bin 2: 21, 21, 24
 - Bin 3: 25, 28, 34
- Smoothing by bin means:
 - Bin 1: 9, 9, 9
 - Bin 2: 22, 22, 22
 - Bin 3: 29, 29, 29
- Smoothing by bin boundaries:
 - Bin 1: 4, 4, 15
 - Bin 2: 21, 21, 24
 - Bin 3: 25, 25, 34



Dealing with Noisy and Missing Data

Outliers

- Outliers are a special kind (subset) of noise
 - Values that clearly deviate from the normal behaviour of the rest of the data
 - Values are in any dimension far away from the expected values and neither similar nor similar to others
 - Can either be a true or valid value because of some phenomena or just be based on a typo
 - Shouldn't be removed or filtered without understanding their origin → Outliers may be valuable
- There are numerous methods to detect outliers that can be classified as follows:

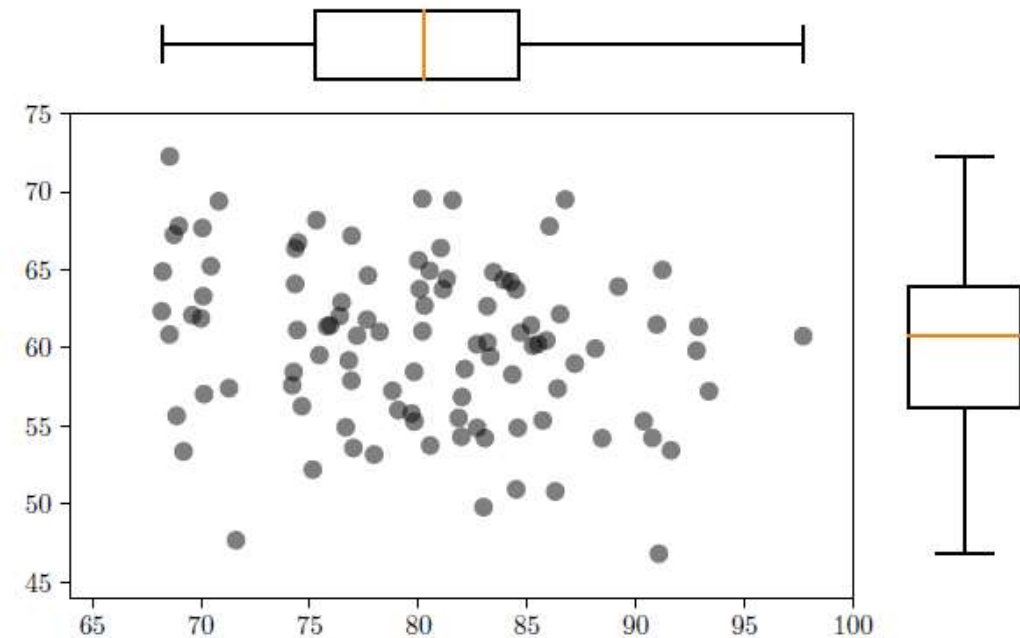


Sources:
Ranga Suri, N. N. R. Outlier detection: Techniques and applications : A data mining perspective. Intelligent systems reference library. 2019

Dealing with Noisy and Missing Data

Statistical Methods to Detect Outliers

- Statistics might seem to be the most efficient and objective approach to detect outliers
 - If the underlying distribution is known, outliers can be detected
 - Outliers have a low probability of belonging to the underlying model (i.e. outside whiskers)
- Most suitable for univariate analysis
 - Boxplots are commonly used to detect outliers in one dimension
- Multidimensional datasets may cause problems
 - Many individual distributions that overlap in a certain sense from each dimension (see right)



Dealing with Noisy and Missing Data

Dealing with Outliers

- Analyse Outliers
 - Outliers may be the goal of the whole analysis e.g. fraud detection, search for anomalies
- Deleting Outliers
 - If the outliers make a small portion of the dataset which is difficult to understand (and contains no other valuable information) they might be deleted.
- Binning Outliers
 - In this context, binning means creating a new category of attributes just for the outliers e.g. “ ≥ 300 ”
 - Simple method that retains most of the original information of the dataset - Very useful for numeric values
- Imputing Outliers
 - Sometimes, outliers occur because of an input error. If so, they might be imputed.
 - If the actual value can be assumed, an imputation with the most likely one might be considered.
 - When the correct value cannot be evaluated, outliers can be handled like missing values

Agenda

What we will cover today...

- Dealing with Noisy and Missing Data
- **Redundancy and Correlation Analysis**
- Normalization
- Discretization
- Data Reduction



Redundancy and Correlation Analysis

Correlation Coefficient

- Correlation between sets of data is a measure of how well they are related. It shows the linear relationship between two sets of data.
 - Correlation coefficients are used to measure how strong a relationship is between two variables.
- There are several types of correlation coefficient formulas. The most common measure of correlation in stats is the (Sample) Pearson Correlation Coefficient

$$r_{xy} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2}}$$

$$r_{xy} = \frac{\sum x_i y_i - n\bar{x}\bar{y}}{\sqrt{(\sum x_i^2 - n\bar{x}^2)} \sqrt{(\sum y_i^2 - n\bar{y}^2)}}$$

n is sample size

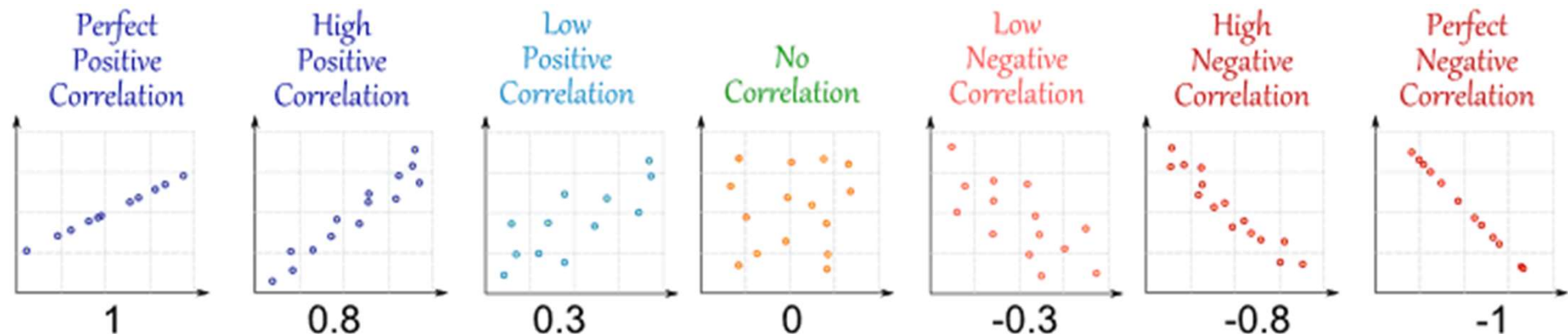
x_i, y_i are the individual sample points indexed with i

$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$ (the sample mean); and analogously for $\bar{y}$

Redundancy and Correlation Analysis

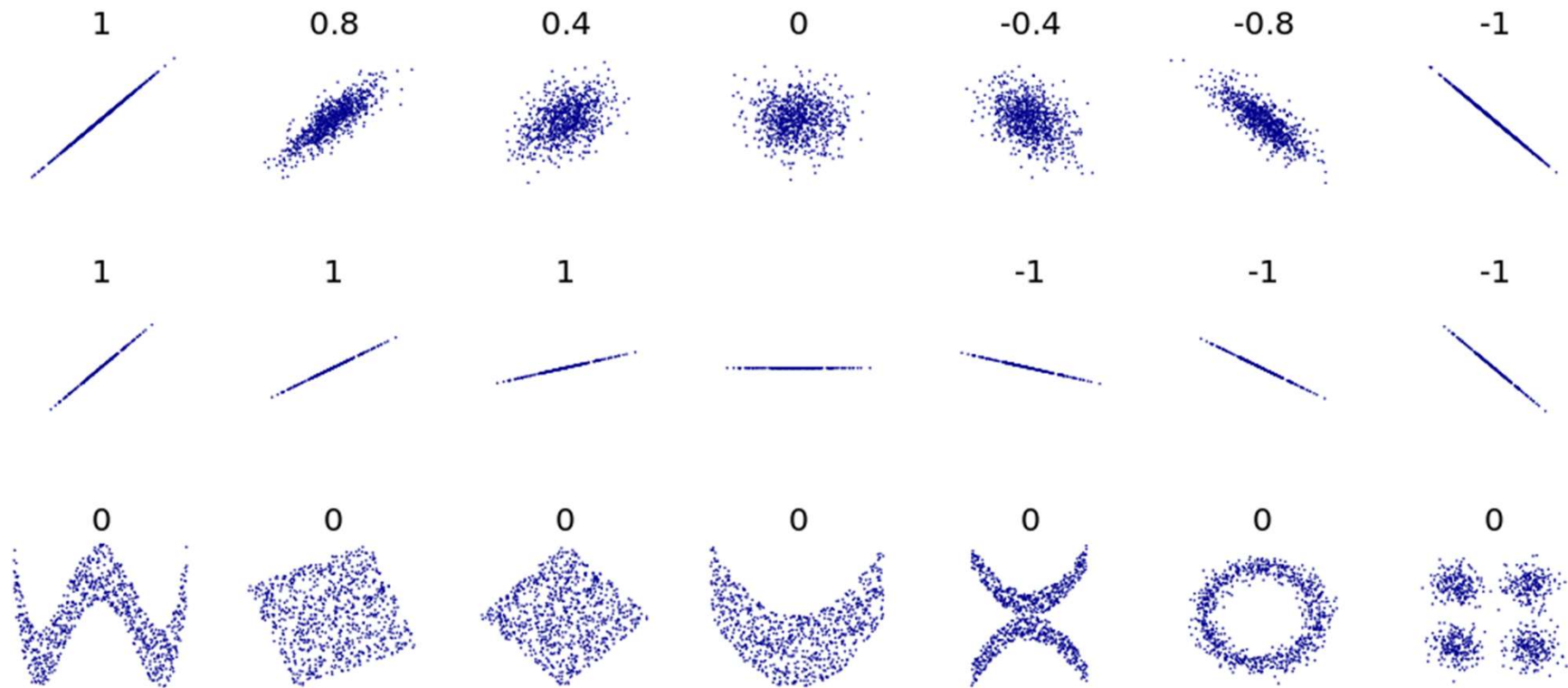
Interpretation of the Correlation Coefficient

- Possible directions
 - Positive Correlation (more → more) e.g.: „more speed, more fuel consumption.“
 - Negative Correlation (more → less) e.g.: „more distance, less fuel in tank.“
- Strength of relationship
 - 0 (=totally independent) to 1 (=strong relationship).

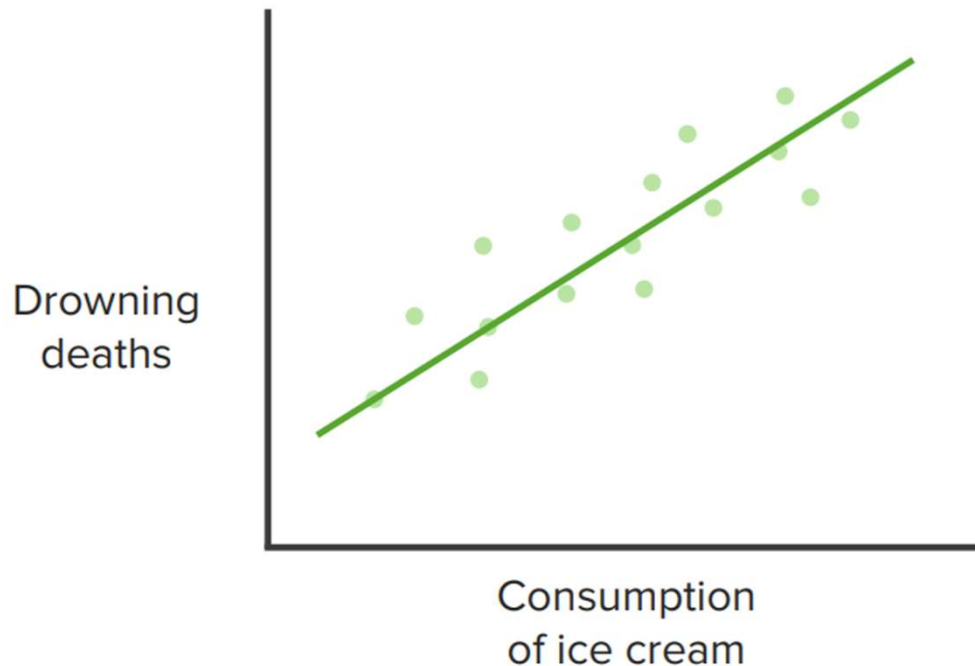


Redundancy and Correlation Analysis

Correlation Examples



Causality vs Correlation



- Causality $\neq$ Correlation
- Causality indicates a logical relationship between two events (cause and effect) as well as a sequence between them (the cause precedes the effect).
- Correlation means that two things are connected, but it does not imply causation.
- Graph: We can see that the consumption of ice cream relates to the amount of drowning deaths. But we can not say that the consumption of ice cream indicates drowning deaths.

Redundancy and Correlation Analysis

Correlation Test for Nominal Data

- χ^2 test (read: chi square) is used for nominal data
 - Suitable for linear and non-linear dependency alike
 - Proof or disproof of the hypothesis that two variables are stochastically independent
 - Calculates the deviation of expected values to observed values = error
 - If sum of errors surmounts a given threshold (the "tolerance value") then there is a dependency
- Calculation of chi square

$$\chi^2 = \sum_{i=1}^c \sum_{j=1}^r \frac{(o_{ij} - e_{ij})^2}{e_{ij}}$$

$$e_{ij} = \frac{\text{count}(A = a_i) \times \text{count}(B = b_j)}{m}$$

Degrees of freedom (df)	Significance level (α)							
	.99	.975	.95	.9	.1	.05	.025	.01
1	-----	0.001	0.004	0.016	2.706	3.841	5.024	6.635
2	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210
3	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.345
4	0.297	0.484	0.711	1.064	7.779	9.488	11.143	13.277
5	0.554	0.831	1.145	1.610	9.236	11.070	12.833	15.086
6	0.872	1.237	1.635	2.204	10.645	12.592	14.449	16.812
7	1.239	1.690	2.167	2.833	12.017	14.067	16.013	18.475
8	1.646	2.180	2.733	3.490	13.362	15.507	17.535	20.090

Redundancy and Correlation Analysis

Chi Square Example

- Given is the following data about students and their lunch behavior:

	WING students observed (expected)	non WING students observed (expected)	sum
OST Mensa	80 (90) (120·150/200)	70 (60) (80·150/200)	150
Elsewhere	40 (30) (120·50/200)	10 (20) (80·50/200)	50
sum	120	80	200

- Calculate a value for chi square:

$$\chi^2 = \frac{(80 - 90)^2}{90} + \frac{(70 - 60)^2}{60} + \frac{(40 - 30)^2}{30} + \frac{(10 - 20)^2}{20} = 11.1$$

- What does this mean?

Agenda

What we will cover today...

- Dealing with Noisy and Missing Data
- Redundancy and Correlation Analysis
- **Normalization**
- Discretization
- Data Reduction



Normalization

Motivation and The Effect of Units

- What happens when we change the unit from kilometer to meter?
 - Changes to smaller units will result to a larger range of for that attribute
 - This may give an attribute an greater effect (or weight)
- Normalization attempts to give all attributes an equal weight
 - The weight of an attribute shall not be dependent on the choice of measurement units
 - Very useful for Classification and Clustering (prevents an outweighing of attributes)
- Normalization transforms the data to fall into a smaller range, usually $[-1, 1]$ or $[0, 1]$
 - There exist several methods to map a value v'_i to alle values v_i of an attribute A
 - Decimal Scaling
 - Min-max Normalization
 - Z-score Normalization

Normalization

Decimal Scaling

- Normalization by decimal scaling simply moves the decimal point of values to an attribute A
 - The number of decimal points moved depends on the maximal absolute value of A
 - A is normalized as follows:

$$v'_i = \frac{v_i}{10^j} \quad \text{where } j \text{ is the smallest integer such that } \max(|v'_i|) < 1$$

- Examples

- $A = [-986, 917] \rightarrow \max(|v_i|) = 986 \rightarrow j = 3 \rightarrow v'_i = \frac{v_i}{1000} \rightarrow A' = [-0.986, 0.917]$
- $A = [-26, 9815] \rightarrow \max(|v_i|) = 9815 \rightarrow j = 4 \rightarrow v'_i = \frac{v_i}{10000} \rightarrow A' = [-0.0026, 0.9815]$
- $A = [43, 99] \rightarrow \max(|v_i|) = 99 \rightarrow j = 2 \rightarrow v'_i = \frac{v_i}{100} \rightarrow A' = [0.43, 0.99]$

Normalization

Min-Max Normalization

- Min-Max Normalization performs a linear transformation on the original data.
 - Minimum min_A and maximum value max_A from the data is fetched, each value is replaced by:

$$v'_i = \frac{v_i - min_A}{max_A - min_A} \quad \text{maps each value in the range } [0,1]$$

- For a predefined range R with $R = [min_R, max_R]$ each value is replaced by:

$$v'_i = \frac{v_i - min_A}{max_A - min_A} \cdot (max_R - min_R) + min_R$$

- Examples

- $A = [120, 980] \rightarrow v_i = 736 \rightarrow R = [0,1] \rightarrow v'_i = \frac{v_i - 120}{980 - 120} \rightarrow v'_i = 0.716$
- $A = [120, 980] \rightarrow v_i = 736 \rightarrow R = [-1,1] \rightarrow v'_i = \frac{v_i - 120}{980 - 120} \cdot (1 + 1) - 1 \rightarrow v'_i = 0.433$

Normalization

Z-Score Normalization

- In this technique, values are normalized based on mean and standard deviation of the data A.
 - Useful when minimum and maximum of A are unknown or there are dominant outliers
 - $\mu, \bar{A}$ and σ, S_A are the mean and the standard deviation, respectively, of attribute A
 - Then the formula is:

$$v'_i = \frac{v_i - \mu}{\sigma} \text{ or } v'_i = \frac{v_i - \bar{A}}{S_A} \text{ also known as z-value}$$

- If A has only few values or there are outliers affecting σ_A the mean absolute deviation S'_A is used instead

$$v'_i = \frac{v_i - \bar{A}}{S'_A} \text{ with } S'_A = \frac{1}{N} (|v_1 - \bar{A}| + |v_2 - \bar{A}| + \dots + |v_N - \bar{A}|)$$

- Example

$$\bar{A}=54, \sigma_A = 16 \rightarrow v_i = 73.6 \rightarrow v'_i = \frac{73.6-54}{16} \rightarrow v'_i = 1.225$$

Agenda

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- **Discretization**
- Data Reduction



Discretization

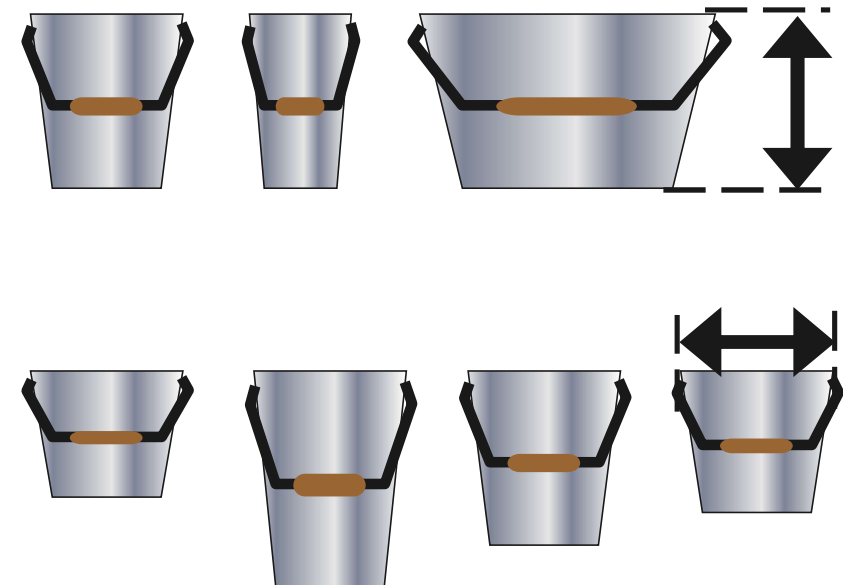
What is Discretization about?

- Discretization is a special form of data reduction
 - Raw data is replaced by a smaller number of interval or concept labels
 - i.e. Transformation of numerical (continuous) values to discrete values
 - Distribute actual values to (few) separate sections (intervals)
 - The interval's value replaces the actual attribute value
- But why?
 - Simplifies the original data and makes mining more efficient
 - Continuous features have a smaller chance of correlating with the target variable
 - Data and results are often easier to understand → better fit to the problem statement
 - There are many algorithms which can only handle nominal attributes (e.g. decision tree algorithms)
 - To eliminate noise in the data

Discretization

Common Discretization Technique: Binning

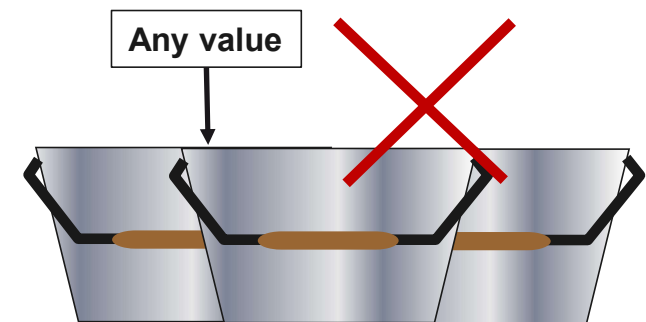
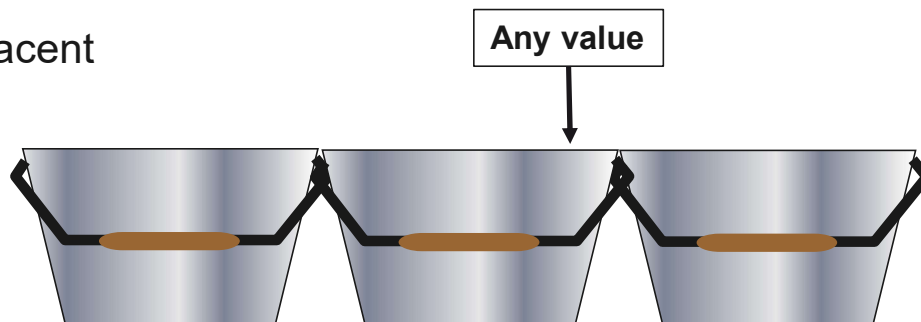
- Numeric attribute values get separated in "bins"
 - Prerequisite: ordered data
 - Number of bins has to be set initially
- bins of equal size ("equidepth binning")
 - Example: 3, 4, 4, 6, 9, 10, 11, 11, 11, 13, 14, 17
 - bin 1: 3, 4, 4, 6
 - bin 2: 9, 10, 11, 11
 - bin 3: 11, 13, 14, 17
- alternative: bins of equal intervals ("equiwidth binning")
 - bin 1: 3, 4, 4, 6
 - bin 2: 9, 10, 11, 11, 11
 - bin 3: 13, 14, 17
- Sensitive to number of bins and outliers



Discretization

Bin boundaries

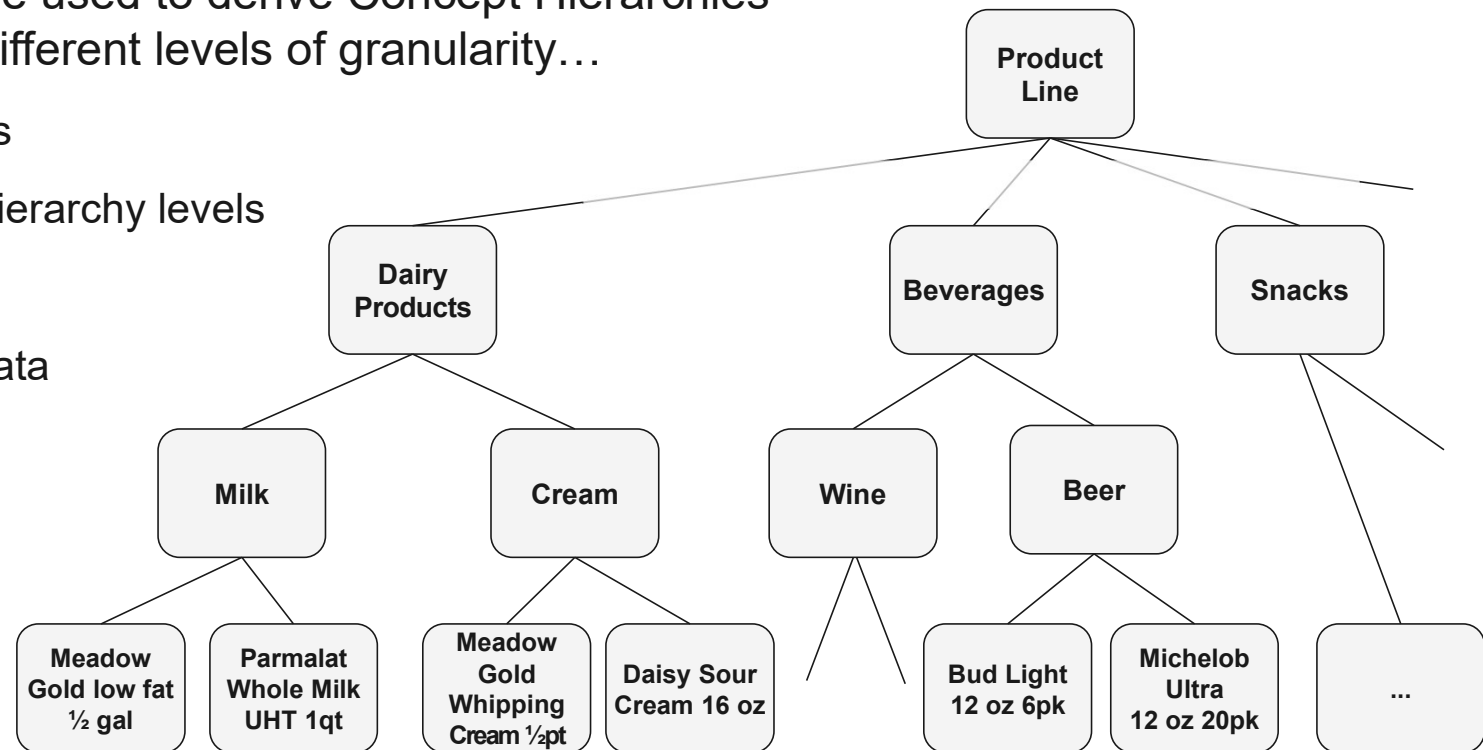
- Recap the final objective of discretization:
 - replace the actual attribute values of an unknown instance with discrete values
 - But: we don't know much about unknown instances...
- Therefore choose bin boundaries
 - to be able to determine any unseen instance later on
 - disjoint and
 - adjacent



Discretization

Concept Hierarchies for Nominal Data

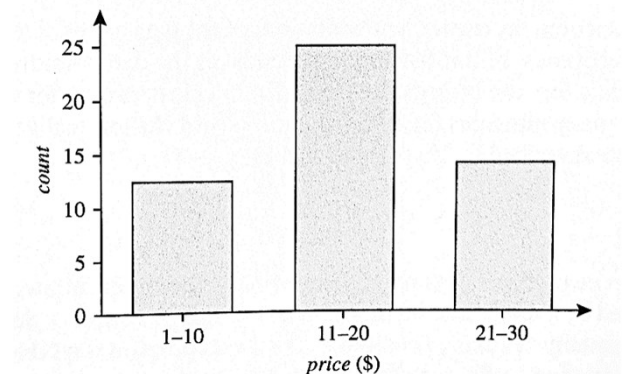
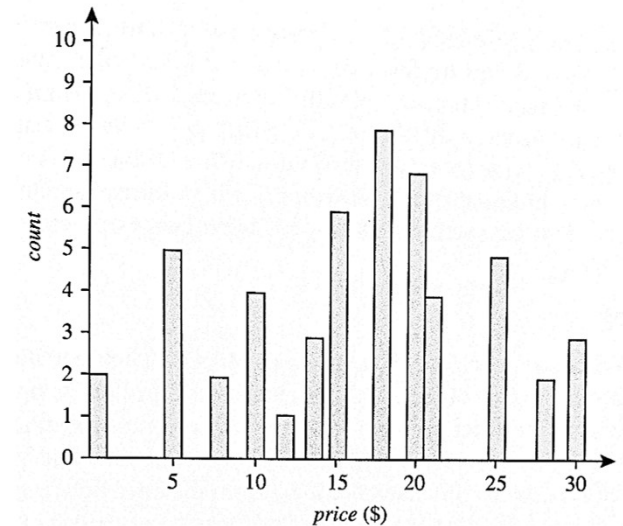
- Discretization may also be used to derive Concept Hierarchies which allows mining on different levels of granularity...
 - Merging several attributes
 - Recursively on multiple hierarchy levels
 - Also useful for smoothing
 - Also useful for numeric data
- Loss of details
- More meaningful
- Manual / Implicit



Discretization

Discretization by Histogram Analysis

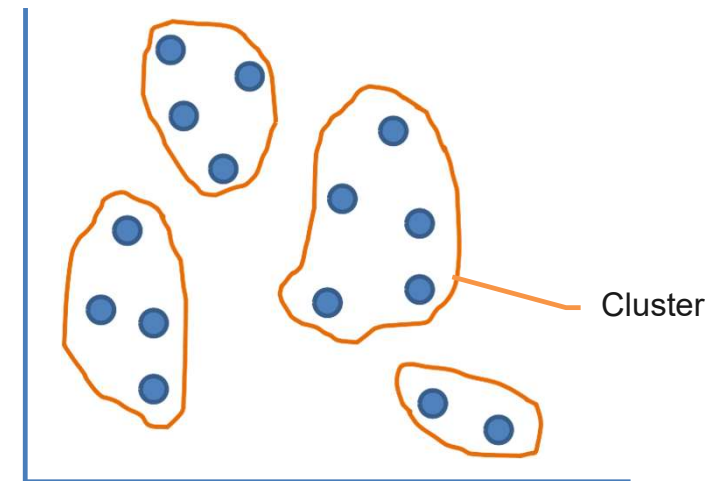
- Recap that a histogram is an approximate representation of the distribution of numerical or categorical data of one attribute.
 - It partitions the values of an attribute in disjoint ranges
- The analysis of an histogram can be applied recursively
 - Generation of a multilevel concept hierarchy
 - Number of concept levels or minimum interval size can be specified



Discretization

Discretization by Clustering

- Clustering is a popular discretization approach for numeric attributes
 - Recap: Clustering Problems → Identify similar (attributes)
 - Takes distribution of attribute A and “closeness” of the datapoints into account
 - High quality discretization results
- Application of a clustering algorithm
 - divide data into segments i.e. partitioning into clusters
 - all values in a segment are replaced by one discrete value (e.g. cluster centers)
- Repeat for further division on sublevels
 - Allows creation of a concept hierarchy
 - Each segment is a node in a concept hierarchy



Agenda

What we will cover today...

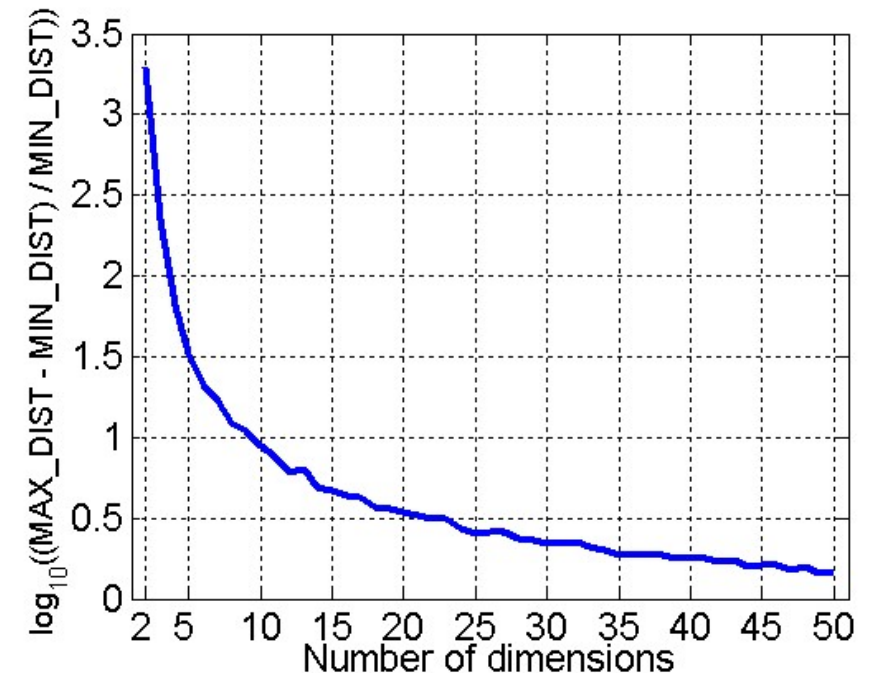
- Dealing with Noisy and Missing Data
- Redundancy and Correlation Analysis
- Normalization
- Discretization
- **Data Reduction**



Data Reduction

Curse of Dimensionality

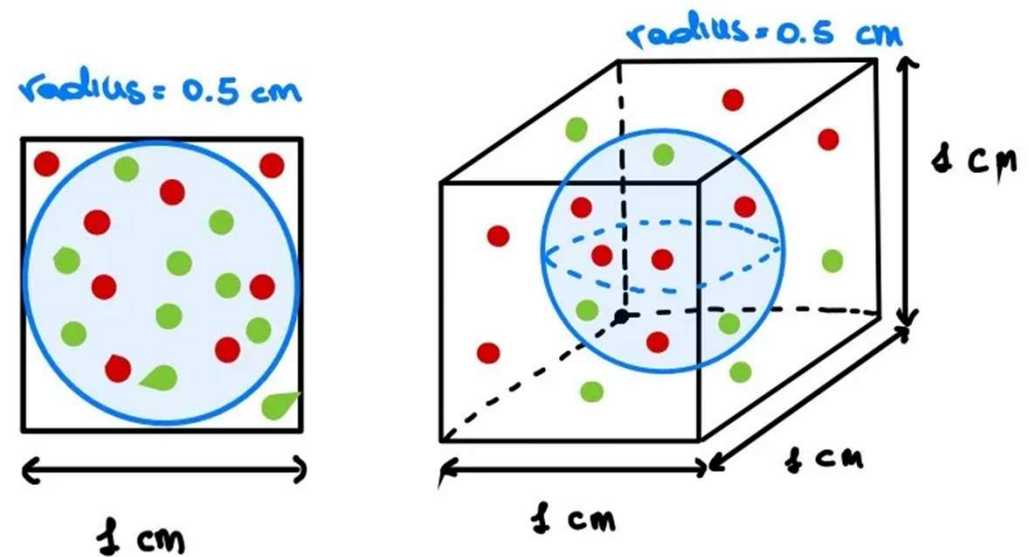
- As the number of features/attributes increases:
 - Volume of feature space increases exponentially.
 - Data becomes increasingly sparse in the space it occupies.
 - Sparsity makes it difficult to achieve statistical significance for many methods.
 - Definitions of density and distance (critical for clustering and other methods) become less useful.
- All distances start to converge to a common value
 - Randomly generate 500 points
 - Compute difference between max and min distance between any pair of points



Data Reduction

Curse of Dimensionality

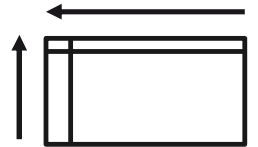
- Example: We have red and green points in our Dataset
- While we have most of the points close to each other at the 2-dimensional space. The Distance increases in the 3-dimensional space.
- In higher-dimensional spaces, most points are located further apart, making distance metrics less reliable
- Consequences: Distance Metrics lose meaning and Algorithms like k-NN and clustering struggle to find meaningful patterns
- Solutions: Dimensionality(Data) reduction, Feature Selection



Data Reduction

Motivation for Data Reduction

- Data Reduction techniques can be applied to obtain a reduced representation of the data set
 - It's mostly about dimensionality (feature) reduction, but also about numerosity reduction
- Why we need it?
 - to improve performance and efficiency (in terms of speed, predictive power, simplicity of the model).
 - to visualize the data for model selection.
 - To reduce dimensionality and remove noise → An issue in times of big data and high speed computing?
- Motivations:
 - removing irrelevant data.
 - increasing predictive accuracy of learned models.
 - improving learning efficiency, such as reducing storage requirements and computational cost
 - reducing the complexity of the resulting model description, better understanding of data and the model.



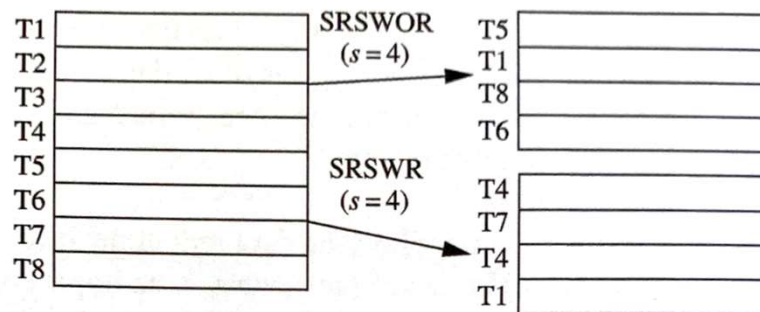
Data Reduction

Sampling to Reduce Numerosity

- Sampling can be used as a data reduction technique because it allows a large data set to be represented by a much smaller random data sample (or subset)
 - There are several methods to draw an sample of size s from a data set D with N tuples
- Simple Random Sample Without Replacement (SRSWOR)
 - Draw s of the N tuples from D . All tuples are equally likely to be sampled.
- Simple Random Sample With Replacement (SRSWR)
 - Similar to the above, but every drawn tuple is placed back into D and may be drawn again.
- Stratified Sample
 - D is divided into mutually disjoint parts called strata
 - A stratified sample is obtained by generating a SRS at each stratum
 - This helps to ensure a representative sample

Data Reduction

Sampling in Pictures



Stratified sample
(according to age)

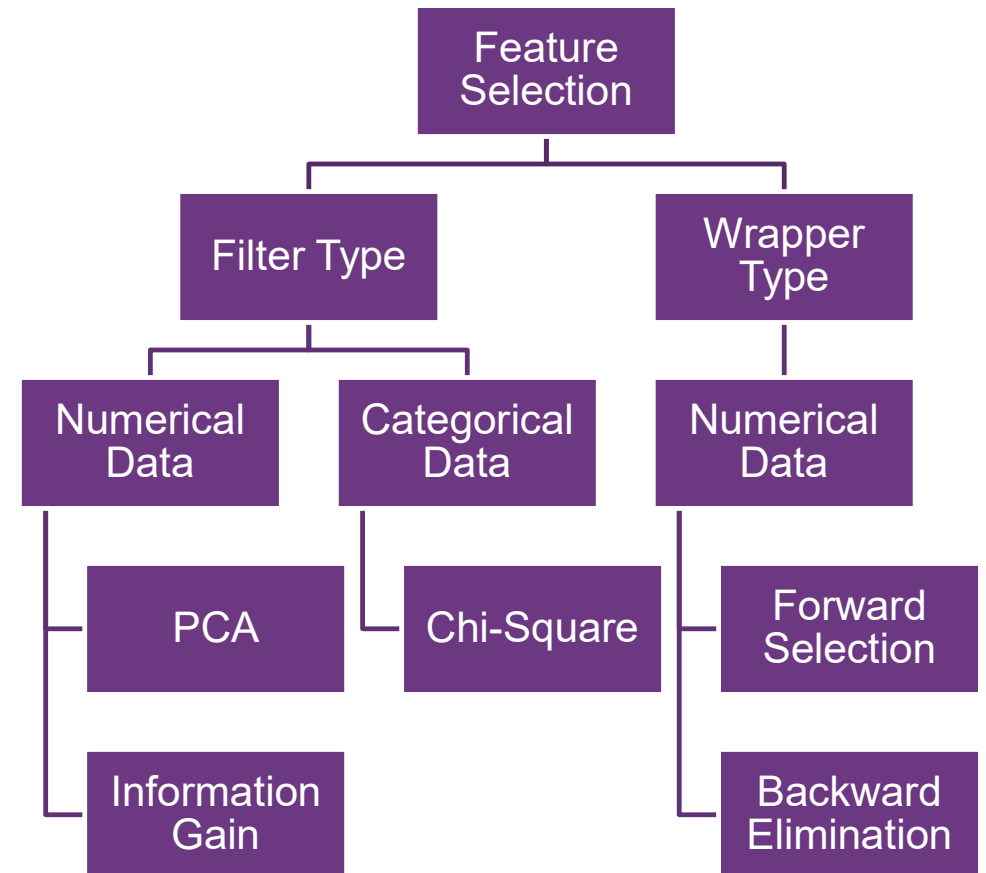
T38	youth
T256	youth
T307	youth
T391	youth
T96	middle_aged
T117	middle_aged
T138	middle_aged
T263	middle_aged
T290	middle_aged
T308	middle_aged
T326	middle_aged
T387	middle_aged
T69	senior
T284	senior

T38	youth
T391	youth
T117	middle_aged
T138	middle_aged
T290	middle_aged
T326	middle_aged
T69	senior

Data Reduction

Übersicht Bekannter Feature Selection Verfahren

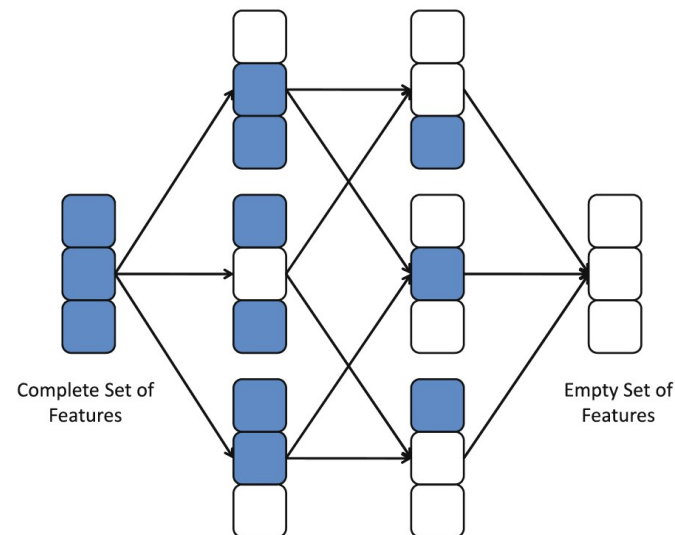
- Filter approaches:
 - Features selected before machine learning algorithm is run
 - “unsupervised”
- Wrapper approaches:
 - Use algorithm as black box to find best subset of features
 - “supervised”
- Both filter and wrapper approaches require:
 - A way to measure the predictive quality of the subset
 - A strategy for searching the possible subsets



Data Reduction

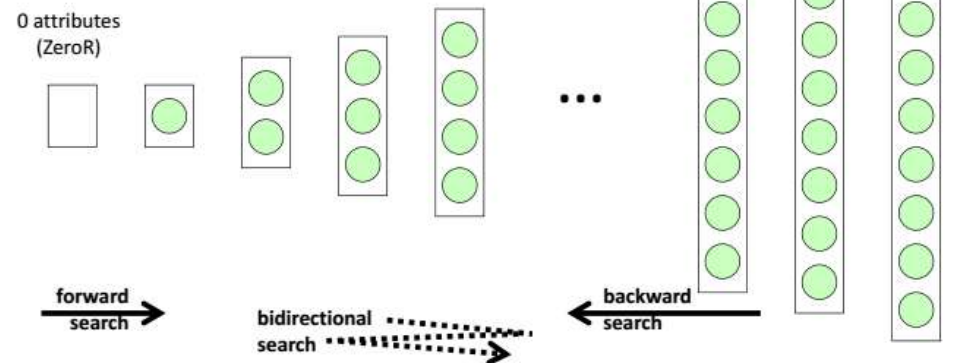
Excursus: Feature Selection is a Search Problem

- Feature Selection can be considered as a search problem
 - Each state of the search space corresponds to a concrete subset of features selected
 - Selection can be represented as a binary array (1 = selected, 0 = not selected)
 - There should be a total of 2^M subsets where M is the number of features of a data set



Searching

- ❖ Exhaustive search: $2^9 = 512$ subsets
- ❖ Searching forward, searching backward
+ when to stop? (*searchTermination*)

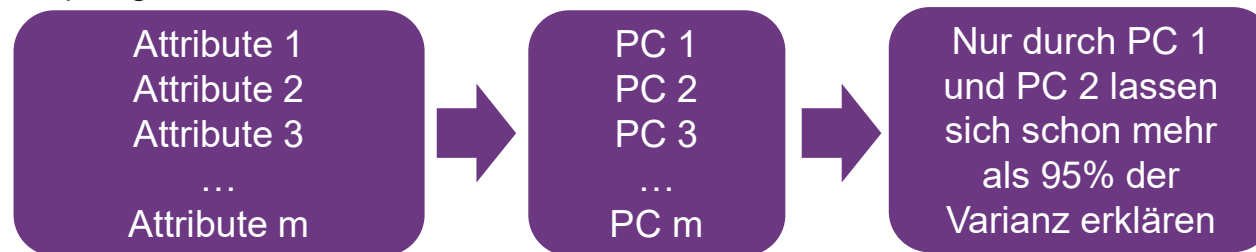


Data Reduction

Excursus: Principal Component Analysis (PCA)

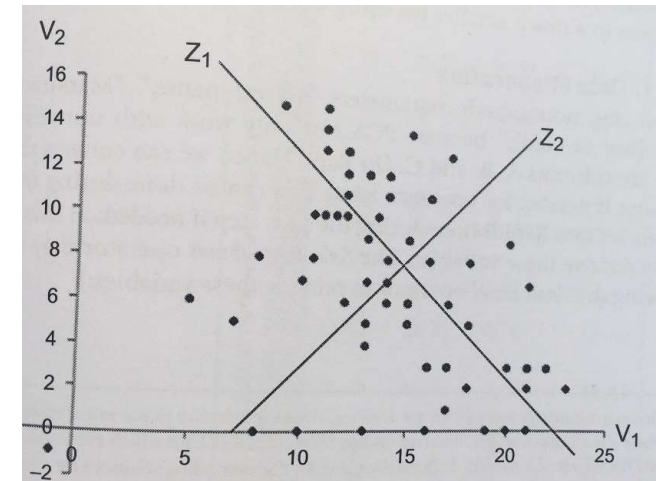
- Pareto als Motivation: Mit 20% der m Variablen, schon 80% der Varianz erklären
 - Principal Components sind «neue voneinander unabhängige Variablen»
 - Principal Components erklären ein Grossteil der Varianz in den Daten
 - Principal Components können mit den ursprünglichen Variablen in Beziehung gesetzt werden

Ursprüngliche Daten:



Transformierte Daten:

$$\begin{aligned} \text{PC 1} &= w_1 \times \text{Attribute 1} + w_4 \times \text{Attribute 4} + w_7 \times \text{Attribute 7} \\ \text{PC 2} &= w_2 \times \text{Attribute 2} + w_5 \times \text{Attribute 5} \end{aligned}$$



Vielen Dank!